DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**EXCHANGE OF TRANSFUSION**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-PICU-012 (v3)**		
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1.0 PURPOSE

1.1. To correct life-threatening electrolyte and fluid imbalance.

1.2. To prevent accumulation of bilirubin in the blood above dangerous level.

1.3. To prevent kernicterus as well as to prevent accumulation of other by hemolysis from hemolytic disease. To replace red blood cells which have poor oxygen releasing capacity and poor carbonic anhydrase activity as in premature infant.

1.4. To remove toxic metabolites and to correct anemia.

2.0 DEFINITION/ABBREVIATION

2.1. **Exchange transfusion** is a technique or procedure used most often to maintain serum bilirubin at levels below neurotoxicity. To exchange all or part of a patient's blood supply for certain medical conditions. A double volume exchange transfusion is replacing the patient's total blood volume twice, leaving the intravascular amount the same. A partial exchange is either increasing or decreasing a patient's hematocrit, while maintaining a constant blood volume. Exchange transfusion is the simultaneous removal of a patient's blood and replacement by donated blood, used in treating serious conditions such as haemolytic disease.

3.0 RESPONSIBILITY

3.1 **Physician** is responsible in assessing patient's condition before and after exchange transfusion and writing an order for any procedure and medication needed.

3.2 **Head Nurse** is responsible to make sure that the unit staffs are well knowledge in assisting transfusion and all needed equipment are available in the unit for emergency cases.

3.3 **Charge Nurses** are responsible to delegate the procedure to a competent nurse before the procedure and to double check physicians' orders if carried out and help the staff nurse throughout the procedure including double checking medications and blood.

3.4 **Staff nurses** are responsible in carrying out physician's orders, preparing needed equipment before the procedure and assisting physician. It is their responsibility as well to frequently reassess patient condition and to maintain sterile technique throughout the procedure and document afterwards.

4.0 POLICY

4.1. Exchange transfusion procedure should be done by a physician under aseptic technique assisted by a nurse.

4.2. Patient should be connected to cardiac monitor and pulse oximeter or continuous vital signs monitoring throughout the procedure.

4.3. Fresh whole blood that is properly typed and cross matched should be used for exchange transfusion.

4.4. Patient should have a central venous line to avoid easy infiltration.

4.5. Blood laboratory pre and Post Exchange for CBC, Hb electrophoresis, Electrolytes, urea and creatinine should be taken.

5.0 PROCEDURE (S)

5.1. Preparatory phase

5.1.1. Correctly identify the patient according to correct patient identification Policy.

5.1.2. Explain the procedure to the patient or parents.

5.1.3. Place infant under radiant warmer and keep his temperature within thermo neutral zone

5.1.4. Attach electronic cardiac monitoring device and pulse oximeter to infant.

5.1.5. Check initial vital signs before procedure for baseline purposes.

5.1.6. Place infant on his back. Restrain all four extremities.

5.1.7. Insert a nasogastric tube if patient has not been NPO.

5.1.8. Check donor blood for type, age and other identifying data.

5.1.9. Wash hands and wear gloves.

5.1.10. Assemble equipment making sure sterility is maintained. Assist the doctor in setting up blood and exchange transfusion equipment.

5.2. Performance phase

5.2.1. Once central venous line is inserted the physician will start exchange transfusion under strict aseptic technique. Document the time exchange transfusion started. Record each successive withdrawals and infusion of blood stating exact amount and time.

5.2.2. After each 100-ml. of blood is exchange, 0.5–1.0 ml. Calcium Gluconate is injected.

5.2.3. In each cycle of blood infusion, 5 ml of normal saline will be given.

5.2.4. After each 100 ml of blood is exchange 0.5-1.0 ml of calcium gluconate is injected. Otherwise, as ordered by physician.

5.2.5. Closely monitor vital signs during procedure.

5.2.6. When the exchange transfusion is completed, Central line may be removed or left in place with an I.V. plug or intravenous Infusion.

5.3. Follow up phase

5.3.1. Document and record the following:

5.3.2. Time transfusion is completed.

5.3.3. Total amount withdrawn and infused Medications given.

5.3.4. Patient's response to procedure Complications.

6.0 EQUIPMENTS

6.1. Cardiac monitor and/or vital signs monitoring machine

6.2. PPE

6.3. Central Venous line

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0017(V2) –Exchange Transfusion– PMSH, 15/2/1439

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

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